GHANA COMMUNICATION TECHNOLOGY UNIVERSITY

COURSE TITLE: AUTOMATA THEORY

COURSE CODE: CSSD 232

NAME OF LECTURER: PRINCE Y. O. AMOAKO

FACULTY: COMPUTING AND INFORMATION SYSTEMS (FoCIS)

DEPARTMENT: COMPUTER SCIENCE

DATE OF SUBMISSION: 8TH JULY 2025

NAME: SEPENU MICHELE ELIKPLIM

INDEX NUMBER: 1699565034

(PROBLEM 2.3)

States: A, B

Proof:
$\delta(A,w)=B \Leftrightarrow$ w contains an odd number of 1's

1. $w = \epsilon$ because:

Reason: ϵ is the only string of length 0
Classification: (C) Reasoning about properties of binary strings

2. $\delta(A, \epsilon) = A$

Reason: By definition of extended transition function
Classification: (B) Reasoning about properties of deterministic finite automata

3. ϵ has an even number of 1's

Reason: ϵ contains 0 ones, and 0 is even
Classification: (C) Reasoning about properties of binary strings

4. Two cases:
   (a) w=x1w = x1w=x1
   (b) w=x0w = x0w=x0

Reason: Every binary string ends in 0 or 1
Classification: (C) Reasoning about properties of binary strings

5. w has an odd number of 1's ⇔ xxx has an even number of 1's
   Reason: One extra 1 flips parity

Classification: (C) Reasoning about properties of binary strings

6. $\delta(A,x)=A \Leftrightarrow x$ has an even number of 1's

Reason: By inductive hypothesis

Classification: (A) Use of the Inductive hypothesis

7. $\delta(A,w)=B \Leftrightarrow w$ has an odd number of 1's $\delta(A, w) = \delta(\delta(A, x),1) = \delta(A,1) = B$

Classification: (B) Reasoning about properties of deterministic finite automata

8. w has an odd number of 1's ⇔ xxx has an odd number of 1's

Reason: Appending 0 doesn't change count of 1's

Classification: (C) Reasoning about properties of binary strings

9. $\delta(A,x)= B \Leftrightarrow x$ has an odd number of 1's

Reason: By inductive hypothesis

Classification: (A) Use of the Inductive hypothesis

10. $\delta(A, w) = B \Leftrightarrow w$ has an odd number of 1's $\delta(A, w)= \delta(\delta(A, x), 0) = \delta(B,0) = B$

Classification: (B) Reasoning about properties of deterministic finite automata